Trainings

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/28/28-002-004-tr.html)
- Drain the oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html)
- Remove the oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
- Remove the oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html)

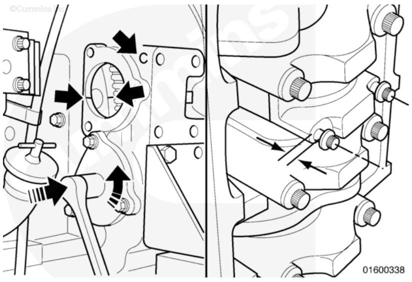
- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Disconnect the air/fuel supply to the air starter, if equipped. Refer to Procedure 012-022 in Section 12. (/qs3/pubsys2/xml/en/procedures/28/28-012-022-tr-gc.html)
- Drain the oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-037-tr-gc.html)
- Remove the oil filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-013-tr-gc.html)

- Drain the coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr-gc.html)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/28/28-002-004-tr-gc.html)
- Remove the oil pan sub-base. Refer to Procedure 007-053 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-053-tr-gc.html)
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-046-tr-gc.html)

Remove

Use the barring mechanism to rotate the engine. Rotate the crankshaft to position the connecting rod at bottom dead center.

Loosen the rod capscrews until there is approximately 6.0 mm [0.24 in] of clearance between the cap and capscrew heads.

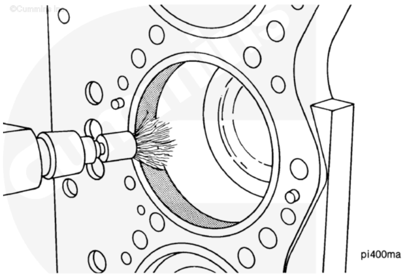


⚠ WARNING ⚠

Wear eye protection. Be sure the wire brush is rated for the rpm being used if the brush is motor driven.

⚠ CAUTION ⚠

To prevent cylinder damage remove any broken wire bristles from the engine.

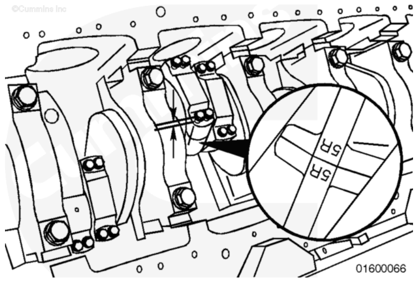


Note : Do not use aluminum oxide cloth to remove the carbon ring unless the cylinder liner will be cleaned thoroughly. Small particles of abrasives in the ring travel area of the cylinder will decrease the durability of the parts.

Use a wire brush. Remove the carbon ring from the top of the cylinder liner. Use a scraper that has an aluminum blade or an abrasive pad, if a wire brush is **not** available.

Note : It is only necessary to remove the amount of the carbon ring required to allow the removal of the pistons.

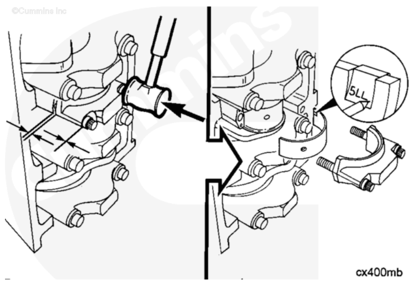
Note : The connecting rod journals in the illustration are positioned at bottom dead center for clarity. If the connecting rod journal is adjacent to a crankshaft counterweight, the rod journal must be at top dead center to remove the rod. The rod will not move past the counter-weight during removal.



The connecting rods **must** have the cylinder number and bank marked on both the rod and the cap on the side positioned toward the camshaft. Check the rods for correct markings. Use a steel stamp to mark any connecting rod that is **not** correctly marked.

Loosen the capscrews until there is 6.0 mm [0.24 in] of clearance between the rod cap and the capscrew head.

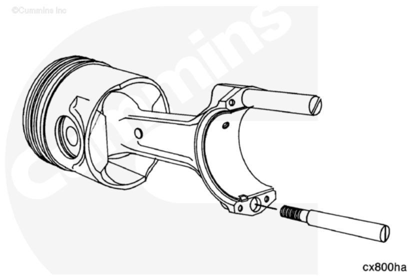
Use a mallet. Tap the connecting rod capscrews until the connecting rod cap and rod separate. Remove the capscrews and the cap.



Remove the lower rod bearing. Use an awl to mark the bearing position in the tang area.

Mark the bearing for future identification or for possible failure analysis.

Use two connecting rod guide pins, Part Number 3375098, or equivalent. Install the guide pins. The guide pins will prevent the rod from falling and damaging the other engine parts.



Push the piston and connecting rod up until the piston rings are above the cylinder liner.

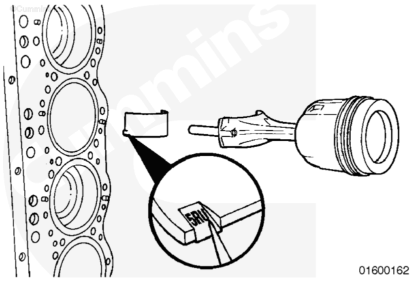
⚠ CAUTION ⚠

Put the piston and rod assembly in a rack to reduce the possibility of damage to the piston and rod assembly.

Remove the piston and connecting rod assembly.

Remove the upper connecting rod bearing.

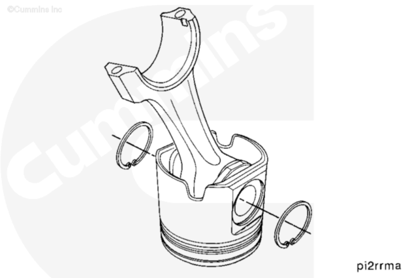
Use an awl to mark the bearing position in the tang area.



Disassemble

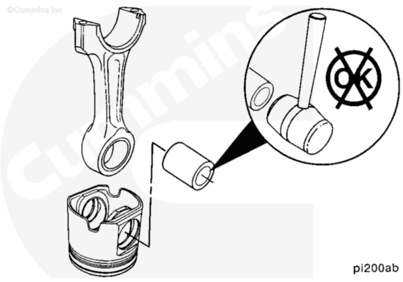
Use internal snap ring pliers to remove the snap rings from both sides of the piston.

Mark or tag the rings from each cylinder with the cylinder number and bank for analysis, if desired.



⚠ CAUTION ⚠

To prevent damage to the piston, do not drop the piston while removing the piston pin.



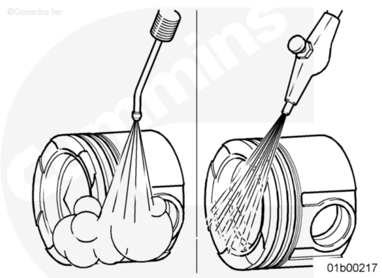
Remove the piston pin from the piston assembly.

The piston and the connecting rod can now be separated.

It is recommended that these parts be kept together for possible failure analysis.

Clean and Inspect for Reuse

Clean and inspect the pistons for reuse. Refer to Procedure 001-043 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-043.html)



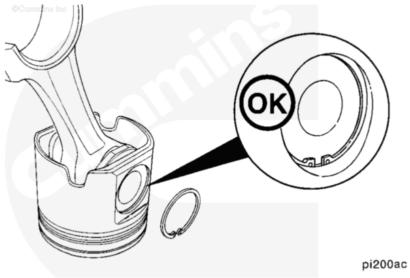
Assemble

Note : In October 1989, a revised premium K piston was released, which removed the requirement to heat the piston prior to fitting on the connecting rod.

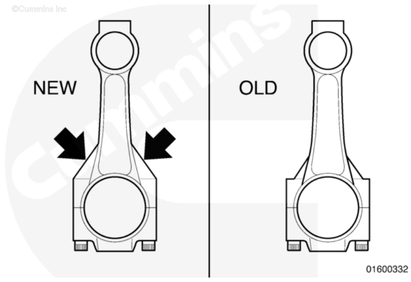
Note : The method described below covers the installation of the revised slip-fit piston.

Install a new snap ring in one piston pin bore of each piston.

It is **not** necessary to heat the pistons before assembly. The piston pin is slip-fit.



There have been two designs of connecting rods for the K38 and K50 engine series. The revised and old connecting rods are interchangeable and can be mixed within an engine.

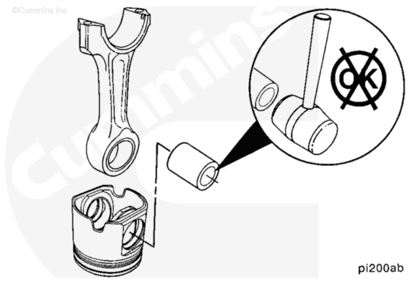


⚠ **CAUTION** ⚠

Do not use a hammer to install the piston pin. The piston can distort, causing it to seize in the liner.

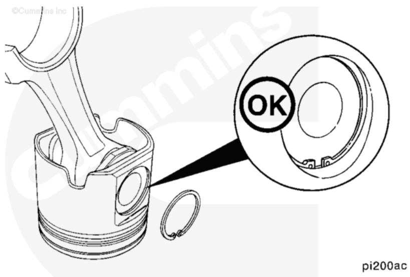
Lubricate the piston pin and connecting rod bushing with clean engine oil.

Align the pin bore of the rod with the pin bore of the piston skirt and crown, and install the piston pin.



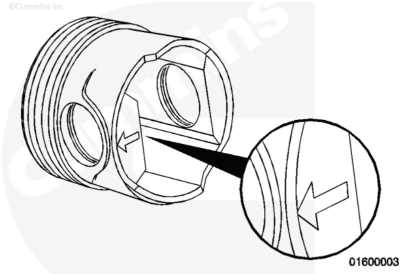
The snap ring **must** be seated completely in the piston groove to prevent engine damage during engine operation.

Install a new snap ring in the piston pin bore.



Offset pin piston on the connecting rod.

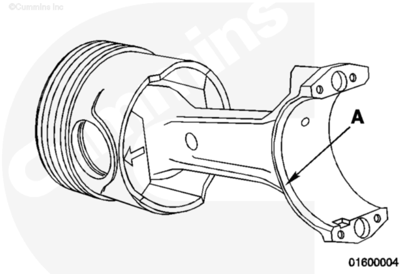
The piston has a cast arrow or the letter F on the bottom of the pin bore. The arrow or the letter F **must** point to or be toward the front of the engine.



Assemble the offset pin with the arrow pointing toward the front of the engine, or the letter F toward the front of the engine.

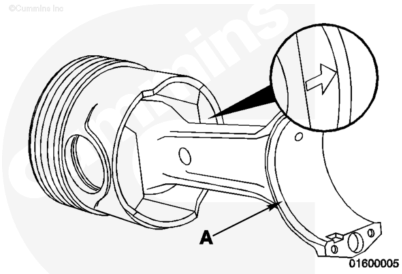
Note : When properly assembled, the arrow or the letter is on the same side as the large chamfer (A) and the lock tang on left bank rods.

Install a new snap ring in the piston pin bore.



Note : When properly assembled, the arrow is on the opposite side of the large chamfer (A) and the lock tang on the right bank rods.

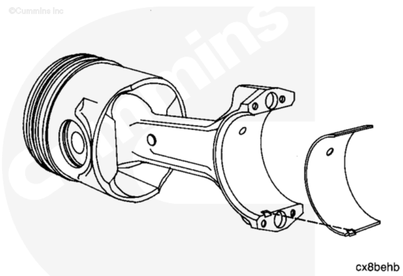
Install a new snap ring in the piston pin bore.



Install

⚠ CAUTION ⚠

The bearings in the same rod and cap must have the same part number. If the numbers are not the same, the bearings can fail and cause severe engine damage.



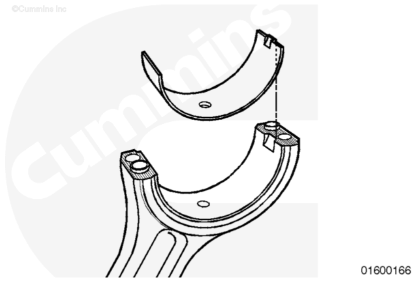
Install the piston cooling nozzles after the piston and connecting rod. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-046-tr.html)

Use a lint-free cloth to clean the connecting rod and the bearing shells.

Install the connecting rod bearing. Make sure the tang is positioned as shown. The end of the bearing **must** be even with the cap mounting surface.

⚠ CAUTION ⚠

To reduce the possibility of engine damage, do not lubricate the backside of the bearing shells.



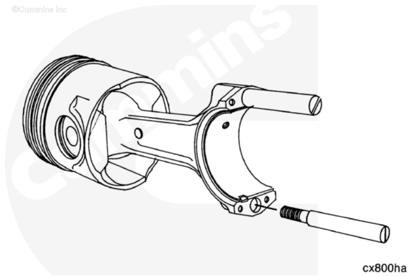
Lubricate the bearing shell surfaces with clean engine oil.

The bearing shells **must** be installed in their original locations, if new bearing shells are **not** used.

Note : All of the connecting rod bearing shells are identical.

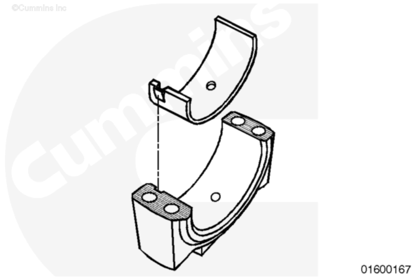
Install two connecting rod guides, Part Number 3375098, or equivalent, in the connecting rod.

The guides will aid the assembly and protect the crankshaft.



⚠ CAUTION ⚠

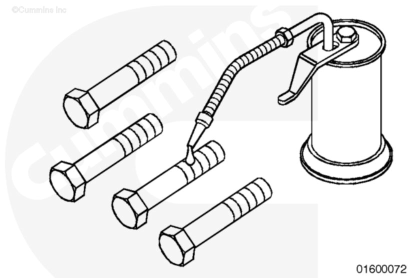
The connecting rods and caps are not interchangeable. The rods and the caps are machined as an assembly. Failure will result if they are mixed.



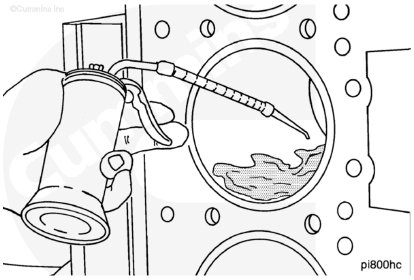
Install the lower bearing shell in the rod cap. Make sure the tang of the bearing shell is in the slot of the cap and the end of the bearing shell is even with the cap surface.

Lubricate the connecting rod capscrews and washers with clean engine oil, as shown.

Install the washers and capscrews in the caps.

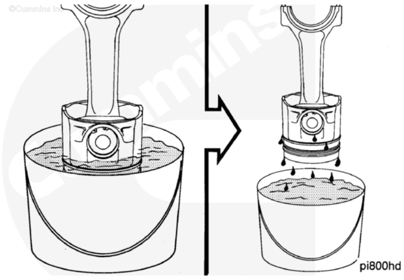


Lubricate the cylinder liner with clean engine oil. The entire bore **must** be lubricated.

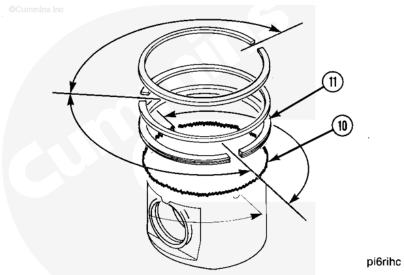


Immerse the piston in engine oil until the rings are covered.

Allow the excess oil to drip off the assembly.

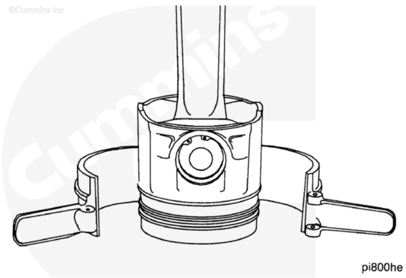


Check to make sure the ring gap position is still correct.



⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure the rings fit correctly in the piston.

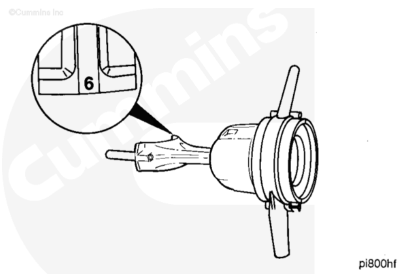


The ring compressor has a tapered bore. The small end of the taper **must** be positioned toward the piston skirt.

Use a piston ring compressor, Part Number 3375342, or equivalent. Install the ring compressor on the piston.

⚠ CAUTION ⚠

The connecting rod and connecting rod cap are a matched set. The cylinder number on the rod and cap must be the same. Damage to the engine can occur if the rods and rod caps are mixed.

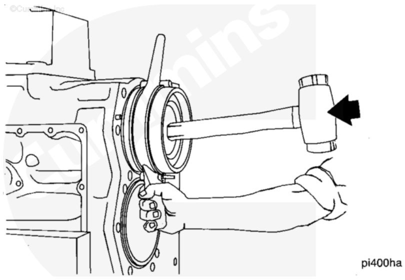


⚠ CAUTION ⚠

The side of the connecting rod with the bearing tang must be on the inboard side of the engine. If the connecting rods are installed backward, engine damage will occur.

Rotate the crankshaft until the journal for the connecting rod being installed is at top dead center and centered in the cylinder bore.

Install the connecting rod and piston until the ring compressor touches the cylinder block.



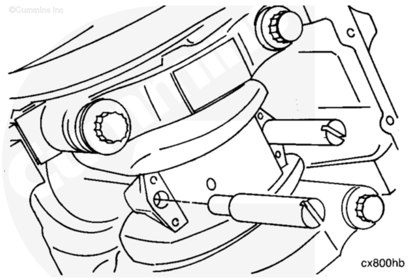
Align the rod with the crankshaft journal.

Hold the ring compressor firmly against the block.

Use a wooden hammer handle to push the piston into the liner.

Note : Verify the arrow on the top of the piston is pointing towards the front (damper end) of the engine.

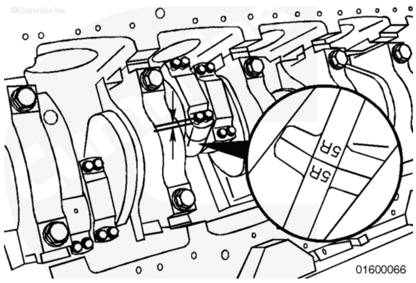
Push the piston into the bore until the rod bearing touches the crankshaft journal.



Remove the guide pins.

⚠ CAUTION ⚠

The connecting rod and connecting rod cap are a matched set. The cylinder number on the rod and cap must be the same. Damage to the engine can occur if the rods and rod caps are mixed.



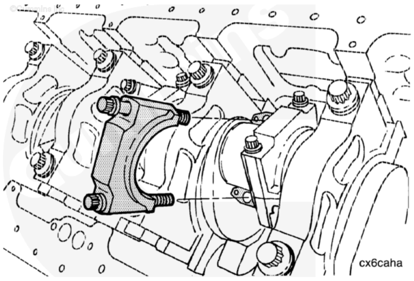
⚠ CAUTION ⚠

The side of the connecting rod with the bearing tang must be on the inboard side of the engine. If the connecting rods are installed backward, engine damage will occur.

Install the connecting rod cap.

⚠ CAUTION ⚠

Do not rotate the crankshaft, the upper rod bearing can fall out. The crankshaft is turned in the illustration for clarity only.



⚠ CAUTION ⚠

The cylinder number on the rod and the cap must be the same. If the numbers are not the same, the parts will fail and cause severe engine damage.

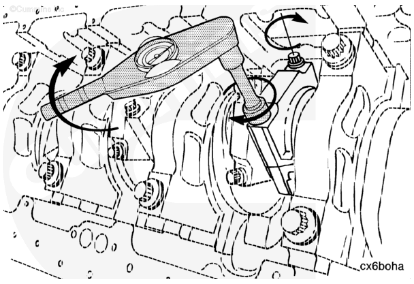
⚠ CAUTION ⚠

The side of the cap with the chamfer must be toward the fillet radius of the crankshaft rod journal. The bearing tang on the rod and cap must be positioned together.

Install the connecting rod cap.

Make sure the numbers match and the numbers are on the same side.

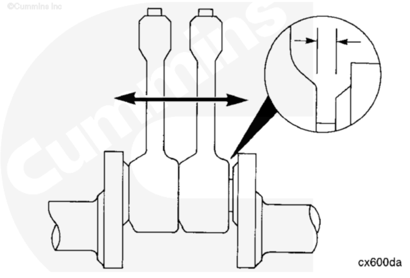
Tighten the bolts alternately and evenly to pull the cap over the dowel pins. Use the following procedure for connecting rod bolt torque sequence. Refer to Procedure 001-005 in Section 1 (/qs3/pubsys2/xml/en/procedures/28/28-001-005-tr.html).



Check the side clearance between the connecting rod and the crankshaft.

The connecting rod **must** move freely from side-to-side.

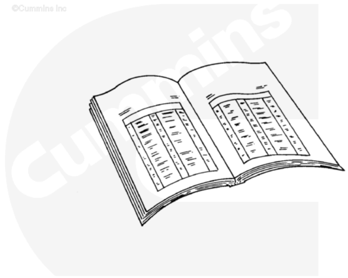
Connecting Rod and Crankshaft Side Clearance		
mm		in
0.41	MIN	0.016
0.56	MAX	0.022



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.


ck800wa

- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/28/28-002-004-tr.html)
- Fill the oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html)
- Install the oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
- Install the oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Operate the engine and check for leaks.

- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-046-tr-gc.html)
- Install the oil pan sub-base. Refer to Procedure 007-053 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-053-tr-gc.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/28/28-002-004-tr-gc.html)
- Install new oil filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-013-tr-gc.html)
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr-gc.html)
- Connect the air/fuel supply to the air starter, if equipped. Refer to Procedure 012-022 in Section 12. (/qs3/pubsys2/xml/en/procedures/28/28-012-022-tr-gc.html)

- Fill the engine with clean engine oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-037-tr-gc.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Perform the engine run-in procedure.

Last Modified: 27-Jan-2023